

Consider the following linear optimization problem:

$$\begin{aligned} & \min_{x \in \mathbb{R}} 2x_1 + x_2 \\ & \text{subject to} \\ & \quad x_1 + x_2 \geq 3, \\ & \quad x_1 + 2x_2 \leq 6, \\ & \quad x_1 \geq 1, \\ & \quad x_2 \geq 1. \end{aligned}$$

1. Does the solution $x_1^* = 1, x_2^* = 2$ verify all the constraints of the problem? Which constraints are active at (x_1^*, x_2^*) ?
2. Answer the same questions with $x_1^* = 2, x_2^* = 1.5$.
3. Answer the same questions with $x_1^* = 1, x_2^* = 1$.